Strategic Data Analysis Project Plan

Project Status: Initiated

Project Lead: Ziad Ibraheem

Final Deadline: December 5, 2026

Primary Objective: End-to-end data pipeline & intelligence

Business Questions Addressed

This project is designed to transform raw UK Train Rides data into actionable intelligence, answering core operational and financial questions:

- **Operational Performance:** What is the On-Time Performance rate compared to delayed/cancelled trips? Which specific routes/stations are most prone to delays, and what are the primary root causes (e.g., signals, weather, technical)?
- **Financial Analysis:** How is revenue distributed across ticket types (Advance, Off-Peak, Anytime)? What is the actual financial loss resulting from refunds tied to delays and cancellations?
- **Passenger Behavior:** When are the peak travel hours and highest-density days? What are the busiest departure and destination stations, and what are the preferred purchasing channels (Online, Kiosks, Desks)?

Project Team & Roles

TEAM LEADER & PYTHON DEVELOPER

Ziad Ibraheem

Oversees project execution, ensures alignment with the final deadline. Handles advanced Exploratory Data Analysis (EDA) and complex Python scripting.

PYTHON DEVELOPER

Hamed Essam

Responsible for data wrangling, handling missing variables, outliers detection, and feature engineering using pandas and NumPy.

SQL DEVELOPER

Eman Abdallah

Executes complex queries, extracts raw data from existing databases, and defines the primary data schemas.

SQL DEVELOPER

Safa Nasser

Optimizes query performance, performs complex data joins, and filters datasets to ensure downstream efficiency.

VISUALIZATIONS (POWER BI)

Ganna Ahmed

Designs the visual architecture, develops complex DAX measures, and ensures charts accurately reflect business logic.

VISUALIZATIONS (POWER BI)

Rehab Maher

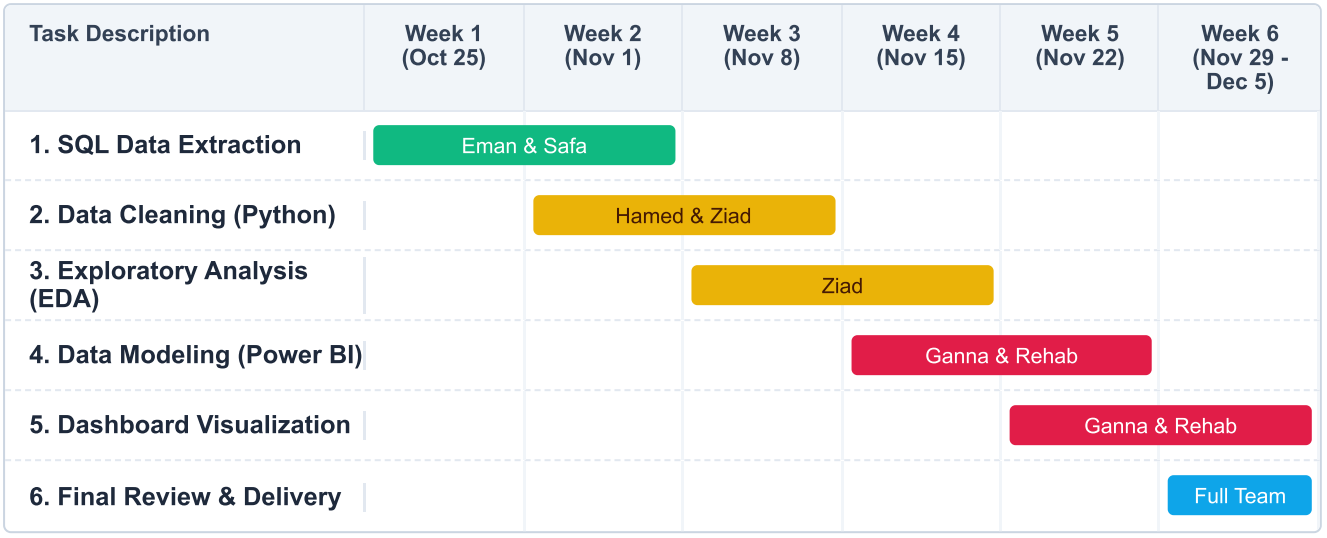
Builds the interactive dashboard interfaces, establishes data modeling relationships within Power BI, and manages report publishing.

Expected Deliverables

- **Structured Relational Database:** Optimized, joined, and filtered datasets ready for analytical querying (SQL layer).
- **Processing Scripts & EDA:** Python scripts documenting data cleaning, missing value handling, and Exploratory Data Analysis.
- **Interactive BI Dashboards:** Comprehensive Power BI reports featuring dynamic filtering, drill-down capabilities, and DAX-driven custom metrics.
- **Executive Insights Report:** A synthesized summary of findings and data-driven recommendations for stakeholders.

Project Timeline (Gantt Chart)

The project spans a 6-week lifecycle, ensuring all dependencies between SQL extraction, Python processing, and Power BI visualization are met before the December 5 deadline.



Key Performance Indicators (KPIs)

- **Total Revenue:** Overall income generated across all ticket categories and routes.
- **On-Time Rate (%):** Percentage of trains arriving exactly on schedule.
- **Cancellation & Delay Rate (%):** Frequency of service disruptions by route.
- **Total Refund Amount:** Direct financial losses due to service failures.
- **Peak Travel Times:** Passenger volume during highest-traffic windows.

Risk Management

- **Data Inconsistency:** Mitigated by strict validation schemas in the SQL phase before handing over to Python.
- **Scope Creep:** Managed by strictly adhering to the defined business questions and avoiding ad-hoc visualization requests.
- **Bottlenecks in Handoffs:** Ziad (Lead) to conduct bi-weekly standups to ensure smooth transitions between SQL, Python, and BI phases.